

# Mathematical and Computational Methods

## Unit 5: Matrices

Unit 4 ended on a promise: the lines and planes we wrote down are the solution sets of *linear systems*, and asking where they meet is asking to solve one. This unit builds the tool. A *matrix* is a rectangular array of numbers that plays two roles at once: it *stores* the coefficients of a linear system, and it *acts* on vectors as a linear transformation—a rotation, a reflection, a stretch. We develop the algebra of matrices, then the two numbers that govern a square matrix—its *determinant* and its *inverse*—and put them to work solving systems and transforming the plane.

**Where this fits.** The determinant met informally in Unit 4—the mnemonic for the cross product, and the triple product that measured volume—is defined here in its own right, and the vectors of Unit 4 are the objects a matrix multiplies and the columns a system is built from. This is the only place in the course where matrices are built from scratch, so the unit is self-contained. Looking ahead, Unit 6 needs all of it: eigenvalues come from a determinant equation,  $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$ ; eigenvectors from solving a system by the elimination developed here; and diagonalisation from an inverse.

## Contents

|                                                               |           |
|---------------------------------------------------------------|-----------|
| Why matrices? A system to return to                           | 2         |
| <b>1 Matrices and their algebra</b>                           | <b>2</b>  |
| 1.1 Definition and notation . . . . .                         | 2         |
| 1.2 Addition, subtraction and scalar multiplication . . . . . | 3         |
| 1.3 Matrix multiplication . . . . .                           | 3         |
| 1.4 Properties, and one that fails . . . . .                  | 4         |
| 1.5 Special matrices . . . . .                                | 4         |
| 1.6 Transpose and trace . . . . .                             | 5         |
| <b>2 The determinant and the inverse</b>                      | <b>6</b>  |
| 2.1 The determinant . . . . .                                 | 6         |
| 2.2 The inverse . . . . .                                     | 7         |
| <b>3 Linear systems and transformations</b>                   | <b>8</b>  |
| 3.1 The inverse-matrix method . . . . .                       | 8         |
| 3.2 Gaussian elimination . . . . .                            | 9         |
| 3.3 What the solutions look like . . . . .                    | 10        |
| 3.4 Matrices as transformations . . . . .                     | 12        |
| <b>Summary</b>                                                | <b>14</b> |

## Learning objectives

By the end of this unit you will be able to:

- ▶ Perform matrix algebra and check multiplication compatibility; use the transpose and trace.
- ▶ Compute determinants and inverses of  $2 \times 2$  and  $3 \times 3$  matrices by hand.
- ▶ Solve linear systems by the inverse method and by Gaussian elimination.
- ▶ Interpret solution sets geometrically and apply transformation matrices.

## Why matrices? A system to return to

Consider the three equations in three unknowns

$$\begin{aligned} 2x + y - z &= 8, \\ -3x - y + 2z &= -11, \\ -2x + y + 2z &= -3. \end{aligned}$$

With two unknowns one substitutes and eliminates by hand without much trouble; with three it is already fiddly and error-prone, and the systems that arise in practice—balancing a chemical equation, finding the currents in a circuit from Kirchhoff’s laws, fitting a model to data—routinely have hundreds or thousands of unknowns. What we want is not a lucky trick but a *systematic, mechanical* procedure that always works and that a computer can carry out. That procedure is what this unit builds. The whole system collapses into the single statement

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix},$$

once we have said what the array  $\mathbf{A}$  is and how it multiplies the column  $\mathbf{x}$ . We build that machinery now and return, in Section 3, to solve this system two different ways.

*Remark (Matrices are how data is stored).* Beyond solving systems, the same rectangular array is the everyday object of data science. A data set is a matrix  $\mathbf{X}$  whose rows are samples and whose columns are features—each row a data point of the kind met in Unit 4. A *transition matrix*, whose columns are probability distributions over states, drives the Markov-chain and PageRank models of Unit 6. Fitting a straight line or a model to data leads to a linear system  $\mathbf{Ax} = \mathbf{b}$  solved by exactly the methods below. So the operations of this unit are not only the engine of physics and engineering calculations but also the basic grammar of machine learning.

## 1 Matrices and their algebra

### 1.1 Definition and notation

**Definition 1.1 (Matrix).** An  $m \times n$  matrix is a rectangular array of (real, and later complex)

numbers with  $m$  rows and  $n$  columns,

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij}),$$

where  $a_{ij}$  is the *entry* in row  $i$  and column  $j$ . The pair  $m \times n$  is its *size* (or *order*). Two matrices are *equal* when they have the same size and equal entries throughout.

A matrix with one column ( $n = 1$ ) is a *column vector*, the objects of Unit 4 written vertically; a matrix with one row is a *row vector*. When  $m = n$  the matrix is *square* of order  $n$ , and its entries  $a_{11}, a_{22}, \dots, a_{nn}$  form the *main diagonal*.

## 1.2 Addition, subtraction and scalar multiplication

These are performed entrywise, exactly as for vectors, and are defined only for matrices of the *same* size.

**Definition 1.2 (The entrywise operations).** For  $\mathbf{A} = (a_{ij})$  and  $\mathbf{B} = (b_{ij})$  of the same size and a scalar  $k$ ,

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij}), \quad k\mathbf{A} = (k a_{ij}), \quad \mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}.$$

**Example 1.1 (Entrywise arithmetic).**

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 4 & -1 \\ 5 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 5 & 5 \end{pmatrix}, \quad 3 \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 0 & 9 \end{pmatrix}.$$

## 1.3 Matrix multiplication

Multiplication is the operation that makes matrices powerful, and it is *not* entrywise. The rule is built so that a linear system becomes  $\mathbf{Ax} = \mathbf{b}$  (shown in the note below), and so that composing two transformations corresponds to multiplying their matrices—a payoff made precise once transformations are defined in §3.4.

**Definition 1.3 (Matrix product).** If  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times p$ —so that the number of *columns* of  $\mathbf{A}$  equals the number of *rows* of  $\mathbf{B}$ —then the product  $\mathbf{AB}$  is the  $m \times p$  matrix whose  $(i, j)$  entry is the dot product of row  $i$  of  $\mathbf{A}$  with column  $j$  of  $\mathbf{B}$ :

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

**Note (The compatibility rule).** The product  $\mathbf{AB}$  exists only when the inner sizes match, and the outer sizes give the result:

$$(m \times \underbrace{n}_{\text{must agree}}) (\underbrace{n}_{\text{must agree}} \times p) \longrightarrow m \times p.$$

If the inner dimensions disagree, the product is undefined.

$$\begin{pmatrix} \boxed{1} & \boxed{2} & \boxed{3} \\ 4 & 5 & 6 \end{pmatrix} \times \begin{pmatrix} \boxed{1} & 0 \\ 0 & \boxed{1} \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \boxed{4} & 5 \\ 10 & 11 \end{pmatrix}$$

$$(\mathbf{AB})_{11} = 1 \cdot 1 + 2 \cdot 0 + 3 \cdot 1 = 4$$

Figure 1: Each entry of  $\mathbf{AB}$  is a row of  $\mathbf{A}$  dotted with a column of  $\mathbf{B}$ ; the shaded row and column produce the shaded entry.

**Example 1.2 (A compatible product).** A  $2 \times 3$  matrix times a  $3 \times 2$  matrix gives a  $2 \times 2$  matrix (Figure 1):

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1+0+3 & 0+2+3 \\ 4+0+6 & 0+5+6 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 10 & 11 \end{pmatrix}.$$

## 1.4 Properties, and one that fails

Matrix multiplication is associative and distributes over addition. Unlike ordinary multiplication, however, it is *not commutative*.

**Theorem 1.1 (Algebraic properties of the product).** Whenever the sizes are compatible,

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}), \quad \mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}, \quad (\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}.$$

In general, however,  $\mathbf{AB} \neq \mathbf{BA}$ .

**Example 1.3 (Order matters).** With  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$ ,

$$\mathbf{AB} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix},$$

so  $\mathbf{AB} \neq \mathbf{BA}$ . The order of multiplication is part of the information.

**Note (A system is a product).** With multiplication defined, the motivating system is literally  $\mathbf{Ax} = \mathbf{b}$ : the  $i$ th entry of  $\mathbf{Ax}$  is row  $i$  of  $\mathbf{A}$  dotted with the column of unknowns, reproducing the  $i$ th equation. Storing a system as a matrix is not just shorthand—it lets the algebra of matrices do the solving.

## 1.5 Special matrices

Certain shapes recur often enough to be named.

**Definition 1.4 (A vocabulary of matrices).**

- The *zero matrix*  $\mathbf{O}$  has every entry 0; it is the additive identity,  $\mathbf{A} + \mathbf{O} = \mathbf{A}$ .
- The *identity matrix*  $\mathbf{I}_n$  is the square matrix with 1s on the main diagonal and 0s elsewhere; it is the multiplicative identity: for a square  $\mathbf{A}$ ,  $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$  (for a rectangular  $\mathbf{A}$  the left and right identities differ in size).
- A *diagonal* matrix has zero off the main diagonal; a matrix is *upper (lower) triangular* if all entries below (above) the diagonal are zero.

$$\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \text{ (diagonal),} \quad \begin{pmatrix} 1 & 4 & 7 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{pmatrix} \text{ (upper triangular).}$$

**1.6 Transpose and trace**

**Definition 1.5 (Transpose).** The *transpose*  $\mathbf{A}^\top$  of an  $m \times n$  matrix  $\mathbf{A}$  is the  $n \times m$  matrix obtained by interchanging rows and columns: the  $(i, j)$  entry of  $\mathbf{A}^\top$  is  $a_{ji}$ . A square matrix with  $\mathbf{A}^\top = \mathbf{A}$  is *symmetric*.

**Example 1.4 (A transpose).**

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad (2 \times 3) \quad \implies \quad \mathbf{A}^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \quad (3 \times 2).$$

The transpose passes through sums unchanged but *reverses the order* of a product—a small fact that will matter repeatedly.

**Proposition 1.1 (Properties of the transpose).**

$$(\mathbf{A}^\top)^\top = \mathbf{A}, \quad (\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top, \quad (k\mathbf{A})^\top = k\mathbf{A}^\top, \quad (\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top.$$

**Definition 1.6 (Trace).** The *trace* of a square matrix is the sum of its diagonal entries,  $\text{tr}(\mathbf{A}) = a_{11} + a_{22} + \cdots + a_{nn}$ .

The trace is linear,  $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$ , and has the useful cyclic property  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$  even though the products themselves differ. Both the trace and the determinant of the next section will reappear in Unit 6 as quantities a matrix carries with it through any change of coordinates.

**Example 1.5 (Trace and the cyclic property).** For  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 4 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\text{tr} \mathbf{A} = 1 + 3 = 4$ . The products differ,  $\mathbf{AB} = \begin{pmatrix} 6 & 1 \\ 3 & 0 \end{pmatrix}$  and  $\mathbf{BA} = \begin{pmatrix} 4 & 11 \\ 1 & 2 \end{pmatrix}$ , yet  $\text{tr}(\mathbf{AB}) = 6 = \text{tr}(\mathbf{BA})$ ,

as the cyclic property promises.

## 2 The determinant and the inverse

To every *square* matrix we attach a single number, the determinant, that decides at a glance whether the matrix can be undone—whether it has an inverse, and hence whether  $\mathbf{Ax} = \mathbf{b}$  has a unique solution.

### 2.1 The determinant

**Definition 2.1 (Determinant of a  $2 \times 2$  matrix).**

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

*Remark* (The determinant is the area and volume of Unit 4). This is no coincidence with the cross product. The number  $ad - bc$  is the signed area of the parallelogram spanned by the columns  $(a, c)$  and  $(b, d)$ —precisely the  $z$ -component of their cross product. In three dimensions the determinant of a matrix is the signed *volume* of the parallelepiped on its columns, the scalar triple product of Unit 4. A determinant of zero therefore means a collapsed, flattened figure—and, as we are about to see, a matrix that cannot be inverted.

For larger matrices the determinant is defined by *cofactor expansion*, reducing an  $n \times n$  determinant to smaller ones.

**Definition 2.2 (Minors, cofactors and expansion).** The *minor*  $M_{ij}$  of a square matrix is the determinant of the submatrix left when row  $i$  and column  $j$  are deleted, and the *cofactor* is the signed minor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

the signs following the checkerboard pattern  $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$ . The determinant is then defined by expansion along the *first* row,

$$\det \mathbf{A} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n},$$

the cofactors  $C_{1j}$  being determinants of smaller matrices, so the definition unwinds down to the  $2 \times 2$  case already given.

*Note* (Expand along whichever row or column is easiest). It is a theorem (proved in the Linear Algebra module) that expanding along *any* row or *any* column gives the same number,

$$\det \mathbf{A} = a_{i1}C_{i1} + \cdots + a_{in}C_{in} = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj},$$

so in practice we pick the row or column with the most zeros. We use this freely below.

**Example 2.1 (A  $3 \times 3$  determinant).** For the coefficient matrix of our system, expand along the first row:

$$\det \mathbf{A} = \begin{vmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{vmatrix} = 2 \begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} + (-1) \begin{vmatrix} -3 & -1 \\ -2 & 1 \end{vmatrix}.$$

The three  $2 \times 2$  determinants are  $(-2 - 2) = -4$ ,  $(-6 + 4) = -2$  and  $(-3 - 2) = -5$ , so

$$\det \mathbf{A} = 2(-4) - 1(-2) + (-1)(-5) = -8 + 2 + 5 = -1.$$

Since  $\det \mathbf{A} \neq 0$ , the matrix is invertible and the system has a unique solution.

## 2.2 The inverse

**Definition 2.3 (Inverse matrix).** A square matrix  $\mathbf{A}$  is *invertible* (or *non-singular*) if there is a matrix  $\mathbf{A}^{-1}$  with

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Such an  $\mathbf{A}^{-1}$  is unique when it exists, and exists precisely when  $\det \mathbf{A} \neq 0$ .

For a  $2 \times 2$  matrix the inverse is immediate: swap the diagonal, negate the off-diagonal, and divide by the determinant.

**Theorem 2.1 (Inverse of a  $2 \times 2$  matrix).** If  $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \neq 0$ , then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

**Example 2.2 (A  $2 \times 2$  inverse).** For  $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$  the determinant is  $6 - 5 = 1$ , so

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix},$$

and indeed  $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$

For a  $3 \times 3$  matrix the same idea works with cofactors: the inverse is the transposed matrix of cofactors—the *adjugate*—divided by the determinant.

*Note* (The cofactor method for  $\mathbf{A}^{-1}$ ).

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \operatorname{adj}(\mathbf{A}), \quad \operatorname{adj}(\mathbf{A}) = \mathbf{C}^T,$$

where  $\mathbf{C} = (C_{ij})$  is the matrix of cofactors. (For larger matrices the row-reduction method of Section 3, applied to  $(\mathbf{A} \mid \mathbf{I})$ , is far quicker; the cofactor formula is the hand method for

$3 \times 3$ .)

**Example 2.3** (*A  $3 \times 3$  inverse, with verification*). Invert  $\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix}$ , for which  $\det \mathbf{A} = -1$ .

*Cofactors.* Deleting each row and column and attaching the checkerboard sign gives the nine cofactors

$$\mathbf{C} = \begin{pmatrix} -4 & 2 & -5 \\ -3 & 2 & -4 \\ 1 & -1 & 1 \end{pmatrix},$$

for instance  $C_{11} = + \begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} = -4$  and  $C_{12} = - \begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} = +2$ .

*Adjugate and inverse.* Transpose  $\mathbf{C}$  and divide by  $\det \mathbf{A} = -1$ :

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T = \begin{pmatrix} -4 & -3 & 1 \\ 2 & 2 & -1 \\ -5 & -4 & 1 \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{-1} \text{adj}(\mathbf{A}) = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix}.$$

*Verification.* Multiplying confirms the inverse:

$$\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbf{I}.$$

We keep  $\mathbf{A}^{-1}$  for the next section, where it solves the system in one stroke.

### 3 Linear systems and transformations

We can now keep the promise of the opening. A linear system is  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , and there are two routes to its solution: invert  $\mathbf{A}$ , or eliminate.

#### 3.1 The inverse-matrix method

If  $\mathbf{A}$  is invertible, multiplying  $\mathbf{A}\mathbf{x} = \mathbf{b}$  on the left by  $\mathbf{A}^{-1}$  isolates  $\mathbf{x}$ :

$$\mathbf{A}\mathbf{x} = \mathbf{b} \implies \mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \implies \mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

**Example 3.1** (*The system by the inverse*). With  $\mathbf{A}^{-1}$  from Section 2 and  $\mathbf{b} = (8, -11, -3)^T$ ,

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix} \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix} = \begin{pmatrix} 32 - 33 + 3 \\ -16 + 22 - 3 \\ 40 - 44 + 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

So  $x = 2$ ,  $y = 3$ ,  $z = -1$ . The method is elegant, but it makes us compute a whole inverse to extract one solution vector. When the goal is simply to solve the system, elimination does

less work.

### 3.2 Gaussian elimination

Elimination works directly on the *augmented matrix*  $(\mathbf{A} \mid \mathbf{b})$ , simplifying it with operations that never change the solution set.

**Definition 3.1 (Elementary row operations).** Each of the following turns a system into an equivalent one:

1. swap two rows;
2. multiply a row by a nonzero scalar;
3. add a multiple of one row to another.

A matrix is in *row echelon form* when each leading (leftmost nonzero) entry sits to the right of the one above it and rows of zeros are at the foot; clearing the entries *above* each leading entry as well, and scaling each to 1, gives *reduced row echelon form*.

*Gaussian elimination* reduces  $(\mathbf{A} \mid \mathbf{b})$  to a triangular (echelon) form and then *back-substitutes*; carrying the reduction all the way to reduced echelon form (*Gauss–Jordan elimination*) lets the solution be read straight off.

**Example 3.2 (The system by Gaussian elimination).** Reduce the augmented matrix, using operations that keep the entries integers:

$$\begin{pmatrix} 2 & 1 & -1 & \mid & 8 \\ -3 & -1 & 2 & \mid & -11 \\ -2 & 1 & 2 & \mid & -3 \end{pmatrix} \xrightarrow[\substack{R_2 \rightarrow 2R_2 + 3R_1 \\ R_3 \rightarrow R_3 + R_1}]{} \begin{pmatrix} 2 & 1 & -1 & \mid & 8 \\ 0 & 1 & 1 & \mid & 2 \\ 0 & 2 & 1 & \mid & 5 \end{pmatrix}$$

$$\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{pmatrix} 2 & 1 & -1 & \mid & 8 \\ 0 & 1 & 1 & \mid & 2 \\ 0 & 0 & -1 & \mid & 1 \end{pmatrix}.$$

The matrix is now triangular. The last row reads  $-z = 1$ , so  $z = -1$ ; the second row gives  $y + z = 2$ , so  $y = 3$ ; and the first gives  $2x + y - z = 8$ , that is  $2x + 3 + 1 = 8$ , so  $x = 2$ . The solution is

$$(x, y, z) = (2, 3, -1),$$

exactly the inverse-matrix answer—two roads to the same destination, and this one needed no inverse. (Gauss–Jordan would continue, clearing the upper entries and scaling, to reach  $(\mathbf{I} \mid \mathbf{x})$  with the solution displayed directly.)

**Note (Row operations and the determinant).** The three elementary operations have predictable effects on the determinant, which is part of why elimination can be trusted: swapping two rows multiplies det by  $-1$ ; multiplying a row by  $k$  multiplies det by  $k$ ; and adding a multiple of one row to another leaves det unchanged. The reduction just performed illustrates the last two of these—no row swap was needed here. Only the step  $R_2 \rightarrow 2R_2 + 3R_1$  altered the size of the determinant—it scaled row 2 by 2, the added multiple of row 1 contributing nothing—while every other operation (each an add-a-multiple) left it fixed.

The matrix produced is triangular, with determinant the diagonal product  $2 \cdot 1 \cdot (-1) = -2$ ; this equals  $2 \det \mathbf{A}$ , so  $\det \mathbf{A} = -1$ , agreeing with the cofactor expansion of Section 2.

The same row reduction also *inverts* a matrix, delivering on the promise of the note in Section 2.2. To find  $\mathbf{A}^{-1}$ , reduce the augmented matrix  $(\mathbf{A} \mid \mathbf{I})$  all the way to  $(\mathbf{I} \mid \mathbf{A}^{-1})$ : the very operations that drive  $\mathbf{A}$  to the identity on the left carry the identity to  $\mathbf{A}^{-1}$  on the right.

**Example 3.3 (The inverse by row reduction).** We invert the running matrix a third way, using the same opening operations as the elimination above. First reduce the left block to upper-triangular form:

$$\left( \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ -3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 + R_1]{R_2 \rightarrow 2R_2 + 3R_1} \left( \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 1 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left( \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 0 & -1 & -5 & -4 & 1 \end{array} \right).$$

Now continue (Gauss–Jordan) by scaling row 3 and clearing the third column above it:

$$\xrightarrow{R_3 \rightarrow -R_3} \left( \begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 3 & 2 & 0 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array} \right) \xrightarrow[R_1 \rightarrow R_1 + R_3]{R_2 \rightarrow R_2 - R_3} \left( \begin{array}{ccc|ccc} 2 & 1 & 0 & 6 & 4 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array} \right).$$

Finally clear the second column of row 1 and scale row 1 to a leading 1:

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left( \begin{array}{ccc|ccc} 2 & 0 & 0 & 8 & 6 & -2 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 3 & -1 \\ 0 & 1 & 0 & -2 & -2 & 1 \\ 0 & 0 & 1 & 5 & 4 & -1 \end{array} \right).$$

The left block is now  $\mathbf{I}$ , so the right block is the inverse,

$$\mathbf{A}^{-1} = \begin{pmatrix} 4 & 3 & -1 \\ -2 & -2 & 1 \\ 5 & 4 & -1 \end{pmatrix},$$

*exactly* the matrix found by cofactors in Section 2—a third road to the same destination, and the one that scales to large systems where cofactors become hopeless.

### 3.3 What the solutions look like

A single linear equation in two unknowns is a line, and in three unknowns a plane (Unit 4). Solving a system asks where these meet, and there are exactly three possibilities, visible already for two lines in the plane (Figure 2).

**Note (Reading the echelon form).** Elimination tells the three cases apart automatically, provided the rows are read in the right order. Look *first* for a row  $(0 \ 0 \ \cdots \ 0 \mid c)$  with  $c \neq 0$ : it reads  $0 = c$ , an impossibility, and settles the matter—*no* solution, whatever the other rows do. If there is no such row the system is consistent, and then a full set of leading entries (one per unknown, as in our example) gives a *unique* solution, while a row of zeros  $(0 \ 0 \ \cdots \ 0 \mid 0)$  leaves an unknown free to take any value: *infinitely many* solutions. For a square system the

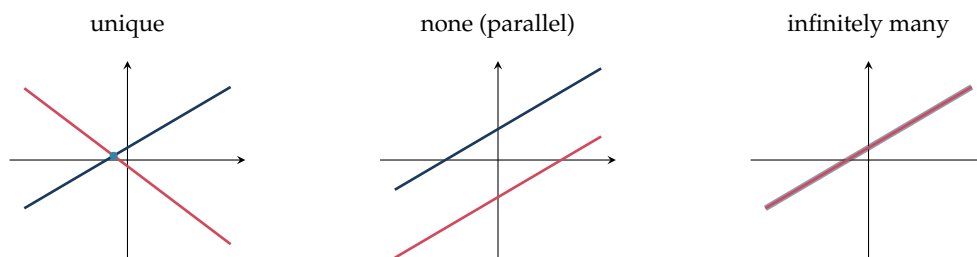


Figure 2: Two lines meet in one point (unique solution), never (no solution, parallel), or everywhere (infinitely many, the same line). Three planes give the same three *counts*—though for planes the no-solution case can also occur with no two of them parallel (they bound a triangular prism).

determinant settles the unique case in advance:  $\det \mathbf{A} \neq 0$  guarantees exactly one solution.

**Example 3.4 (A system with infinitely many solutions).** Not every consistent system pins down a single point. Consider

$$\begin{aligned}x + y + z &= 6, \\x + 2y + 3z &= 14, \\2x + 3y + 4z &= 20.\end{aligned}$$

Reducing the augmented matrix with the same elimination as before,

$$\begin{aligned}\left(\begin{array}{ccc|c}1 & 1 & 1 & 6 \\1 & 2 & 3 & 14 \\2 & 3 & 4 & 20\end{array}\right) &\xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{ccc|c}1 & 1 & 1 & 6 \\0 & 1 & 2 & 8 \\0 & 1 & 2 & 8\end{array}\right) \\&\xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c}1 & 1 & 1 & 6 \\0 & 1 & 2 & 8 \\0 & 0 & 0 & 0\end{array}\right).\end{aligned}$$

The last row has become all zeros: the third equation was the sum of the first two and carried no new information, so one unknown is left *free*. Set  $z = t$ . The second row then gives  $y + 2z = 8$ , so  $y = 8 - 2t$ , and the first gives  $x = 6 - y - z = -2 + t$ . The solution is a whole one-parameter family,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R},$$

which is exactly the vector equation of a *line* from Unit 4: the three planes meet not in a point but along a common line. (Put  $t = 0, 1, 2$  to read off the points  $(-2, 8, 0)$ ,  $(-1, 6, 1)$ ,  $(0, 4, 2)$ , each of which satisfies all three equations.) Had the right-hand side instead produced a row  $(0 \ 0 \ 0 \mid c)$  with  $c \neq 0$ , the very same elimination would have signalled *no* solution.

**Remark** (Looking ahead: rank and the general theory). The number of leading entries left after elimination—the count of genuinely independent equations—is called the *rank* of the matrix, and comparing two of them decides between the three cases for a system of any size: the rank of the coefficient matrix  $\mathbf{A}$  against the rank of the augmented  $(\mathbf{A} \mid \mathbf{b})$ . If

the augmented rank is the larger, some row reads  $0 = c$  and there is no solution; if they agree, the solution is unique when that common rank equals the number of unknowns and a family otherwise. (The coefficient rank alone cannot decide: the exercise sheet keeps the very matrix above, of rank 2, but changes the right-hand side until the system is inconsistent.) We use rank here only informally, through the shape of the echelon form. The systematic theory of rank, of general  $m \times n$  systems, and of abstract vector spaces (independence, basis and dimension, first met geometrically in Unit 4) is developed in the dedicated *Linear Algebra module*, for which this unit is the computational foundation.

### 3.4 Matrices as transformations

A square matrix does more than store a system: multiplying a vector by an  $n \times n$  matrix sends  $\mathbb{R}^n$  to itself, and the map  $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$  is a *linear transformation*—it preserves vector addition and scaling. In the plane, the  $2 \times 2$  matrix is read off by seeing where it sends the basis vectors  $\mathbf{i} = (1, 0)$  and  $\mathbf{j} = (0, 1)$ : those images are its columns.

*Note* (The standard plane transformations).

$$\underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{\text{rotation by } \theta} \quad \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\text{reflect in } y=x} \quad \underbrace{\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}}_{\text{enlargement } \times k} \quad \underbrace{\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}}_{\text{stretch}} \quad \underbrace{\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}}_{\text{shear}}$$

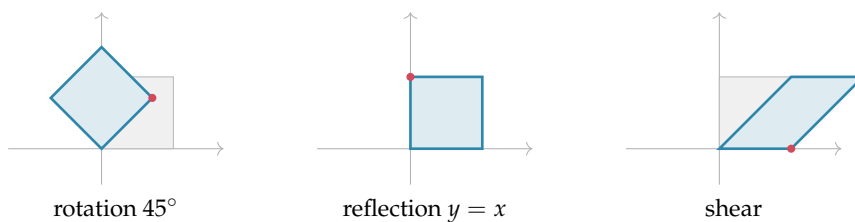


Figure 3: The unit square (grey) under three transformations; the marked corner is the image of  $\mathbf{i}$ , the first column of the matrix.

**Example 3.5 (Composing and reading a transformation).** A rotation by  $90^\circ$  has matrix  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , and it sends  $\mathbf{i} = (1, 0)$  to  $(0, 1)$  and  $\mathbf{j} = (0, 1)$  to  $(-1, 0)$ —a quarter-turn, the real-plane echo of multiplying by  $i$  in Unit 3. Applying one transformation after another corresponds to *multiplying* the matrices, in right-to-left order: doing  $\mathbf{S}$  then  $\mathbf{R}$  is the single matrix  $\mathbf{RS}$ . Because the determinant scales area,  $\det \mathbf{R} = 1$  keeps area and orientation, a reflection has  $\det = -1$  (area kept, orientation flipped), and a singular matrix ( $\det = 0$ ) collapses the plane onto a line—or, in the extreme case of the zero matrix, onto the origin alone.

*Remark* (Translations are not linear). A translation  $\mathbf{x} \mapsto \mathbf{x} + \mathbf{t}$  shifts every point by a fixed vector, but it is *not* a linear transformation: it moves the origin, whereas  $\mathbf{A}\mathbf{0} = \mathbf{0}$  always. Such maps are called *affine*. The standard fix—used throughout computer graphics—is to lift the plane into  $\mathbb{R}^3$  with an extra coordinate 1, after which a single  $3 \times 3$  matrix can carry

out rotation, scaling *and* translation together; but within  $\mathbb{R}^2$ , translation lies outside the reach of a  $2 \times 2$  matrix.

## Summary

| Concept          | Key idea                                                                                                                                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Matrix           | $m \times n$ array $(a_{ij})$ ; entrywise $+$ and scalar $\times$ (same size).                                                                                                                                        |
| Multiplication   | $(\mathbf{AB})_{ij}$ = row $i$ of $\mathbf{A}$ $\cdot$ column $j$ of $\mathbf{B}$ ; needs columns of $\mathbf{A}$ = rows of $\mathbf{B}$ ; $(m \times n)(n \times p) \rightarrow m \times p$ .                        |
| Properties       | Associative and distributive, but <i>not</i> commutative: $\mathbf{AB} \neq \mathbf{BA}$ in general.                                                                                                                  |
| Special          | Zero $\mathbf{O}$ , identity $\mathbf{I}$ ( $\mathbf{AI} = \mathbf{A}$ ), diagonal, triangular.                                                                                                                       |
| Transpose        | Swap rows/columns; $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ ; symmetric if $\mathbf{A}^T = \mathbf{A}$ . Trace $\text{tr } \mathbf{A} = \sum a_{ii}$ .                                                           |
| Determinant      | $2 \times 2$ : $ad - bc$ (signed area). Larger: cofactor expansion $\det \mathbf{A} = \sum_j a_{ij} C_{ij}$ , $C_{ij} = (-1)^{i+j} M_{ij}$ .                                                                          |
| Inverse          | Exists iff $\det \mathbf{A} \neq 0$ ; $2 \times 2$ formula $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ ; $3 \times 3$ via $\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \text{adj } \mathbf{A}$ . |
| System, two ways | $\mathbf{Ax} = \mathbf{b}$ : inverse method $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ , or Gaussian elimination on $(\mathbf{A} \mid \mathbf{b})$ then back-substitute.                                                |
| Solution types   | Unique ( $\det \neq 0$ ), none ( $a$ $0 = c$ row), or infinitely many (a free variable)—meeting in a point, not at all, or along a line/plane.                                                                        |
| Transformations  | $\mathbf{x} \mapsto \mathbf{Ax}$ is linear; columns are the images of $\mathbf{i}, \mathbf{j}$ ; rotation/reflection/stretch/shear; composition = product; $ \det $ = area scale. Translation is affine, not linear.  |

*Next unit: eigensystems—the special directions a matrix merely stretches. Using the determinant for the characteristic equation, elimination for the eigenvectors, and the inverse for diagonalisation, we find the axes along which  $\mathbf{Ax} = \lambda\mathbf{x}$ , and use them to compute matrix powers and the long-run behaviour of evolving systems.*

Companion material: a separate *Exercise Sheet 5* (with solutions) develops the practice problems for this unit.